

Dynamic Pricing and Dispatching for Mobile Charging Services: An MDP-Based Two-stage Sequential RL Framework

1 Background and Research Problem

Emerging mobile charging technologies extend service flexibility for electric vehicle users beyond conventional fixed charging infrastructure. We consider a charging service operator that jointly manages a set of fixed charging stations (FCSs) and a fleet of mobile charging vehicles (MCVs), with the objective of maximizing the overall profit from its fixed and mobile charging services. While prior studies have largely considered MCVs either as mobile charging resources for serving demand or as flexible energy resources for grid support, we allow MCVs to perform both roles. The operator therefore leverages the mobility and operational flexibility of MCVs to provide more accessible and convenient charging services, complement the existing FCS infrastructure, and generate additional revenue through energy discharge. This dual functionality requires the operator to jointly coordinate MCV deployment, operation, and pricing. Existing MCV dispatch studies, however, typically assume exogenously specified demand or predetermined charging prices, thereby overlooking how MCV pricing influence users' charging choices and how pricing decisions should account for users' responses. This motivates our central research question:

Can an MDP-based reinforcement learning (RL) framework effectively learn joint MCV pricing and dispatch policies while accounting for endogenous user charging choices to improve the overall profitability of an integrated FCS-MCV operator?

2 Proposed Approach

We formulate the operator's sequential decision problem as a two-stage hierarchical RL framework. At each period t , the decision process unfolds in two stages, with users' charging choices endogenized within the decision process. The objective is to maximize the long-term profit of the integrated FCS-MCV operation. In the first stage, the operator observes the system state s_t ,

$$s_t = (\{l_{mt}, e_{mt}\}_{m \in \mathcal{M}}, \lambda_t, \mathbf{p}_t^F)$$

where l_{mt} and e_{mt} denote the location and battery energy of MCVs, respectively. λ_t denotes EV charging requests; and $\mathbf{p}_t^F$ denotes FCS charging and discharging prices. The operator then quote price for MCV service:

$$\mathbf{p}_t^M \sim \pi_{\theta_1}^P(\cdot | s_t).$$

where $\pi_{\theta_1}^P$ is the policy network for the first-stage pricing decision. Given the resulting MCV prices, users choose among available MCVs, FCSs, and an outside charging option based on a discrete choice model, inducing endogenous demand:

$$\mathbf{D}_t = \mathcal{D}(\mathbf{p}_t^M; \xi_t),$$

where ξ_t represents exogenous factors affecting charging choices. MCV pricing can therefore attract demand that would otherwise remain unserved, while also potentially shifting demand away from existing FCSs. In the second stage, after user choices are realized, the system transitions to the post-choice state $\tilde{s}_t = (s_t, \mathbf{p}_t^M, \mathbf{D}_t)$. Each MCV $m \in \mathcal{M}$ is modeled as an RL agent, and observes only a local observation $o_{mt} = (\{l_{mt}, e_{mt}\}, \mathbf{p}_t^F, \mathbf{p}_t^M, \mathbf{D}_t)$ of the post-choice state. Based on this observation, each MCV $m \in \mathcal{M}$ independently selects its action according to:

$$a_{mt}^D \sim \pi_{\theta_{2m}}^D(\cdot | o_{mt}).$$

where $\pi_{\theta_{2m}}^D$ denotes the policy network for each MCV. Specifically, MCVs may travel to service location to serve EV demand, reposition across service locations, recharge or discharge at FCS, subject to operational and battery constraints.

The two decision stages are coupled through the following sequential process:

$$s_t \xrightarrow[\text{Stage 1}]{\pi_{\theta_1}^P} \mathbf{p}_t^M \xrightarrow[\text{Stage 2}]{\text{user choice}} \mathbf{D}_t \xrightarrow[\text{Stage 2}]{\pi_{\theta_{2m}}^D, m \in \mathcal{M}} \{\mathbf{a}_{mt}^D\}_{m \in \mathcal{M}} \longrightarrow (r_t, s_{t+1}).$$

where r_t represents the operator's total profit from the integrated FCS-MCV operation, defined as revenues from FCS and MCV charging services and MCV energy discharge, minus net of energy procurement, travel, operating, and unmet-demand costs. The pricing and dispatch policies are jointly trained to maximize the operator's expected discounted profit over the planning horizon.

3 Expected Contributions

1. We develop an MDP based framework that **endogenizes user charging choices**, allowing the model to capture both potential expansion of the operator's served market and demand shifts between FCS and MCV services.
2. We propose a **two-stage RL approach** that jointly learns dynamic MCV pricing and dispatching policies.
3. We conduct numerical experiments comparing the proposed framework with a myopic greedy strategy benchmark to evaluate its effectiveness in terms of profitability and operational performance.